

Homework 3

MATH 346/377 - Winter 2025

Due on Sunday, February 23rd, 9:00pm on Crowdmark

Math 346: Do problems 1 to 6 – Total of points= 60pts

Math 377: Do problems 1 to 7 – Total of points= 70pts

The following Pari GP commands could be useful, in particular for question 1b):

- The command `Mod(a^(b), n)` returns `Mod(r, n)`, where $0 \leq r \leq n - 1$ is the residue $a^b \bmod n$. It also works to find inverses, for example `Mod(2^(-1), 11)` returns `Mod(6, 11)`, since 6 is the inverse of 2 modulo 11.
- The command `factor(n)` returns the prime factorization $[p_1, a_1], \dots, [p_r, a_r]$. For example `factor(1210104)` returns `[2 3] [3 2] [7 5]`, so $1210104 = 2^3 \cdot 3^2 \cdot 7^5$.

The chinese remainder states that if $m_1, \dots, m_r$ are coprime integers, then for any integers $a_1, \dots, a_r$ there is an integer x such that

$$x \equiv a_i \pmod{m_i} \quad i \in \{1, \dots, r\}. \quad (1)$$

Moreover, this integer x is unique modulo $m := m_1 \cdots m_r$. Since the m_i are all coprime, the integer $\frac{m}{m_i}$ is coprime to m_i , and there is an integer b_i such that $b_i \frac{m}{m_i} \equiv 1 \pmod{m_i}$. One can then easily verify that the integer

$$x = a_1 b_1 \frac{m}{m_1} + \cdots + a_r b_r \frac{m}{m_r} \in \mathbb{Z} \quad (2)$$

satisfies (1). Moreover, as the b_i go through the residue class $b_i + m_i \mathbb{Z}$, this gives infinitely many integers satisfying the congruence.

1. (11 points) An integer is *square-free* if it is a product of distinct primes¹, otherwise it is said to be *non square-free*.
 - (a) (5 points) Show that for each integer k , we can find k consecutive integers that are non square-free. *Hint: Use the Chinese Remainder Theorem*

¹So the only exponents appearing in the prime factorizations are 0 and 1

- (b) (6 points) For $k = 3$ we have for example the following three consecutive non square-free integers $48 = 3 \cdot 2^4$, $49 = 7^2$ and $50 = 2 \cdot 5^2$. Use (2) to find a sequence of four consecutive non square free-integers. Use Pari GP to write down the prime factorization of these numbers and verify that they are non-square free.

Hint: The numbers will probably get large so you want to use Pari GP here to help you with computations.

Solution:

- (a) If we can find a sequence of k consecutive non square-free integers, then it means we can find an integer x such that $x - k, x - k - 1, \dots, x - 2, x - 1$ are each divisible by some p_i^2 , where p_i is a prime. Equivalently, this means that $x \equiv i \pmod{p_i^2}$. Hence, to find such a sequence we can start by taking k distinct primes $p_1, \dots, p_k$. Since the primes are coprime, by the Chinese Remainder theorem we can find an x such that $x \equiv i \pmod{p_i^2}$ for $1 \leq i \leq k$. This shows that $x - k, x - k - 1, \dots, x - 2, x - 1$ are all non square-free.

- (b) We fix four primes 2, 3, 5, 7 and we find to find x such that

$$\begin{aligned} x &\equiv 1 \pmod{2^2 = 4}, \\ x &\equiv 2 \pmod{3^2 = 9}, \\ x &\equiv 3 \pmod{5^2 = 25}, \\ x &\equiv 4 \pmod{7^2 = 49}. \end{aligned}$$

So in the notation of question 1, we have $m_1 = 4, m_2 = 9, m_3 = 25, m_4 = 49$ and $m = 4 \cdot 9 \cdot 25 \cdot 49 = 44100$. We also have

$$\begin{aligned} \frac{m}{m_1} &= \frac{44100}{4} = 11025 \equiv 1 \pmod{4}, \quad \text{so the inverse is } b_1 \equiv 1 \pmod{4}, \\ \frac{m}{m_2} &= \frac{44100}{9} = 4900 \equiv 4 \pmod{9}, \quad \text{so the inverse is } b_2 \equiv 7 \pmod{9}, \\ \frac{m}{m_3} &= \frac{44100}{25} = 1764 \equiv 14 \pmod{25}, \quad \text{so the inverse is } b_3 \equiv 9 \pmod{25}, \\ \frac{m}{m_4} &= \frac{44100}{49} = 900 \equiv 18 \pmod{49}, \quad \text{so the inverse is } b_4 \equiv 30 \pmod{49}. \end{aligned}$$

We deduce that

$$\begin{aligned} x &= a_1 b_1 \frac{m}{m_1} + a_2 b_2 \frac{m}{m_2} + a_3 b_3 \frac{m}{m_3} + a_4 b_4 \frac{m}{m_4} \\ &= 11025 + 2 \cdot 7 \cdot 4900 + 3 \cdot 9 \cdot 1764 + 4 \cdot 30 \cdot 900 = 235253. \end{aligned}$$

So one sequence is given by

$$x - 4 = 235249 = 7^2 \cdot 4801 \tag{3}$$

$$x - 3 = 235250 = 2 \cdot 5^3 \cdot 941 \tag{4}$$

$$x - 2 = 235251 = 3^3 \cdot 8713 \tag{5}$$

$$x - 1 = 235252 = 2^2 \cdot 103 \cdot 571. \tag{6}$$

Note that the next integer $235253 = 43 \cdot 5471$ is squarefree. On the other hand, before we have $235248 = 2^4 \cdot 3 \cdot 13^2 \cdot 29$ which is also non square-free, but $235247 = 367 \cdot 641$. So in fact we have found a sequence of five consecutive non square-free integers.

2. (6 points) Show that the equation $37x^2 + 2 = y^{18}$ has no solutions in the integers.

Solution: If the equation had a solution in the integers, it would have a solution modulo any integer n . Taking $p = 37$ we find the equation

$$2 \equiv y^{18} \pmod{37}. \quad (7)$$

Now y cannot be divisible by 37, so any solution y must be coprime to 37. By Fermat's little theorem, this would imply that $1 \equiv y^{36} = (y^{18})^2 \equiv 2^2 \equiv 4 \pmod{37}$. Hence there is no solution modulo 37, and thus no solution in the integers.

3. (12 points) Let $\phi(n)$ be Euler's ϕ -function.
 (a) (6 points) Show that if d divides n , then $\phi(d)$ divides $\phi(n)$.
 (b) (6 points) Find all $n \geq 1$ for which $\phi(n) = 16$.

Solution:

(a) We have

$$\phi(n) = n \prod_{p|n} \left(1 - \frac{1}{p}\right), \quad \phi(d) = d \prod_{p|d} \frac{p-1}{p}, \quad (8)$$

so that

$$\frac{\phi(n)}{\phi(d)} = m \prod_{\substack{p|n \\ p \nmid d}} \frac{p-1}{p} \quad (9)$$

where $n = md$. Now any prime p that divides n but not d must divide m . Hence, we have

$$\frac{\phi(n)}{\phi(d)} = m \prod_{p|m} \frac{p-1}{p} \quad (10)$$

is an integer, since $\prod_{p|m} p$ divides m .

(b) Let $n = \prod_{p|n} p^{a_p}$ be the prime factorization of n . Then

$$\phi(n) = \prod_{p|n} p^{a_p-1}(p-1) = 16. \quad (11)$$

Hence the primes p that can appear must satisfy p or $p-1$ divides 16. This gives a finite list of possibilities: $p = 2, 3, 5, 17$. So n must be of the form $2^a 3^b 5^c 17^d$. Note that if 17 appears in the prime factorization of n , then 16 appears in the prime factorization of $\phi(n)$. Hence, either $n = 17$ or $n = 2^a 3^b 5^c$. For the remaining cases, we see that

$$\phi(2^a 3^b 5^c 17^d) = 2^{a-1} \cdot 3^{b-1} \cdot 2 \cdot 5^{c-1} \cdot 4 \cdot 17^{d-1} 16 = 16. \quad (12)$$

implies that $0 \leq b, c, d \leq 1$. Moreover, if $d = 1$ then $b = c = 0$. We get 6 different cases:

$$\begin{aligned} 16 &= \phi(17), \\ 16 &= \phi(2 \cdot 17), \\ 16 &= \phi(2^a) = 2^{a-1}, \text{ which implies } a = 5, \\ 16 &= \phi(2^a \cdot 3) = 2^{a-1} 2, \text{ which implies } a = 4, \\ 16 &= \phi(2^a \cdot 5) = 2^{a-1} 4, \text{ which implies } a = 3, \\ 16 &= \phi(2^a \cdot 3 \cdot 5) = 2^{a-1} 8, \text{ which implies } a = 2. \end{aligned}$$

So the six solutions to $\phi(n) = 16$ are

$$n \in \{17, 32, 34, 40, 48, 60\}. \quad (13)$$

4. (10 points) Find all the roots of $x^{18} + 4x^{14} + 3x + 10 \equiv 0 \pmod{21}$.
Verify your solutions with *Pari GP*.

Solution: By the Chinese Remainder theorem, this equation has a solution mod $21 = 3 \cdot 7$ if and only if it has a solution modulo 3 and 7.

First modulo 7. Clearly, we cannot have $x \equiv 0 \pmod{7}$, since otherwise we would have $10 \equiv 0 \pmod{7}$. So x is coprime to 7 and we have $x^6 \equiv 1 \pmod{7}$. Hence, for x coprime to 7 we also have $x^{18} \equiv 1 \pmod{7}$ and $x^{14} \equiv x^2 \pmod{7}$. The equation reduces to

$$x^{18} + 4x^{14} + 3x + 10 \equiv 4x^2 + 3x + 4 \equiv 0 \pmod{7}, \quad (14)$$

when x is coprime to 7. The discriminant of this quadratic equation is $\Delta = 3^2 - 4 \cdot 4 \cdot 4 \equiv 1 \pmod{7}$. Hence we have $\Delta = r^2$ with $r = 1$, the discriminant is a quadratic residue and we have two solutions modulo 7:

$$\frac{-b - r}{2a} = \frac{-3 - 1}{8} \equiv -4 \equiv 3 \pmod{7} \quad (15)$$

and

$$\frac{-b + r}{2a} = \frac{-3 + 1}{8} \equiv -2 \equiv 5 \pmod{7}. \quad (16)$$

Looking at the equation modulo 3 we also find that x must be coprime to 3, and by Fermat's little theorem we have $x^2 \equiv 1 \pmod{3}$. Hence we have $x^{18} \equiv x^{14} \pmod{3}$, and

$$x^{18} + 4x^{14} + 3x + 10 \equiv 1 + 4 + 0 + 1 \equiv 0 \pmod{3} \quad (17)$$

for any x coprime to 3. Hence, any $x \equiv 1, 2 \pmod{3}$ satisfies the equation.

In conclusion, the four roots of the equation are the residue classes $x \pmod{21}$ such that $x \equiv 3, 5 \pmod{7}$ and $x \equiv 1, 2 \pmod{3}$:

$$x \in \{5, 10, 17, 19\}. \quad (18)$$

5. (10 points) What are the last two digits of $3^{3^{58}}$? Justify your answer.

Solution: We can find the last two digits of a number by taking its residue modulo 100. (For example $123456 = 100 \cdot 1234 + 56$.)

So we have to compute $3^{3^{58}} \pmod{100}$. By Fermat's little theorem, we have that $x^{\phi(n)} \equiv 1 \pmod{n}$ if x is coprime to n . Since 3 is coprime to 100, if we know that $3^{58} \equiv r \pmod{\phi(100) = 40}$, then $3^{3^{58}} \equiv 3^r \pmod{100}$. On the other hand, if we know that $5^8 \equiv s \pmod{\phi(40) = 16}$, then $3^{5^8} \equiv 3^s \pmod{100}$ since 3 is coprime to 40. So we are left with computing $5^8 \pmod{\phi(40) = 16}$. Since $\phi(16) = 8$ and 5 is coprime to 16, we have that $5^8 \equiv 1 \pmod{16}$. It follows that

$$3^{5^8} \equiv 3 \pmod{\phi(100) = 40}, \quad (19)$$

and thus

$$3^{3^{58}} \equiv 3^3 \equiv 27 \pmod{100}. \quad (20)$$

In conclusion, the last two digits of $3^{3^{58}}$ are 27.

6. (a) (6 points) Let x be an integer coprime to n . For $i \geq 1$, show that x and x^i have the same order if and only if i is coprime to the order of x .
- (b) (5 points) Find all residues of order 14 modulo $p = 43$. Justify your answer.

Solution:

- (a) Let d be the order of x modulo n , and let r be the order of x^i .

First suppose that i is coprime to d . We want to show that $r = d$. On the one hand, we have

$$1 \equiv (x^i)^r \equiv x^{ir} \pmod{n}, \quad (21)$$

which implies that d divides ir . But since i and d are coprime, d divides r . On the other hand, since

$$1 \equiv 1^i \equiv (x^d)^i \equiv (x^i)^d \pmod{n}, \quad (22)$$

we also have that r divides d . So $r = d$.

Conversely, assume that i and d are not coprime. We want to show that $r \neq d$. Let us denote by $g := \gcd(i, d)$ the gcd between i and d . We have

$$(x^i)^{\frac{d}{g}} \equiv (x^d)^{\frac{i}{g}} \equiv 1 \pmod{n}, \quad (23)$$

so that r divides $\frac{d}{g}$. In particular, we cannot have $r = d$.

- (b) We know that there are $\phi(14) = 6$ residues of order 14. If we find one residue a of order 14, then the other residues of order 14 are a^i with i coprime to 14:

$$a, a^3, a^5, a^9, a^{11}, a^{13}. \quad (24)$$

By a result we have seen in class, x has order 14 if and only if $x^{14} \equiv 1 \pmod{43}$ and $x^{\frac{14}{q}} \not\equiv 1 \pmod{43}$ for all primes dividing 14 (so $q = 2$ and $q = 7$). We can use this to check that 2 has order 14, since

$$2^2 \equiv 4 \pmod{43}, \quad 2^7 \equiv -1 \pmod{43}, \quad 2^{14} \equiv 1 \pmod{43}. \quad (25)$$

The other residues of order 14 are

$$\{2, 2^3, 2^5, 2^9, 2^{11}, 2^{13}\} \equiv \{2, 8, 22, 27, 32, 39\} \pmod{43}. \quad (26)$$

7. (10 points) Show that if p and q are primes, then the cyclotomic polynomial

$$\Phi_q(x) = 1 + x + \cdots + x^{q-1} \quad (27)$$

has $q - 1$ distinct roots modulo p if $p \not\equiv 1 \pmod{q}$, has just one root if $p = q$, and has none otherwise.

Solution: First note that

$$x^q - 1 = (x - 1)(1 + x + x^2 + \cdots + x^{q-1}) = (x - 1)\Phi_q(x). \quad (28)$$

hence, any root of Φ_q must be a root of $x^q - 1$.

If $p - 1 \not\equiv 0 \pmod{q}$, then $(q - 1, p) = 1$ and we can find $a, b \in \mathbb{Z}$ such that $aq + b(p - 1) = 1$. Hence, if $x^q \equiv 1 \pmod{p}$ then $1 \equiv (x^q)^a \equiv x^{1-b(p-1)} \equiv x \pmod{p}$. Hence, if $p - 1 \not\equiv 0 \pmod{q}$ then the only possible root of $x^q - 1$ (and thus of $\Phi_q(x)$) is 1. We have $\Phi_q(1) = q$, so 1 is a root if $q = p$, and not a root otherwise.

Now suppose that q divides $p - 1$, say $p - 1 = qm$. Then $x^q - 1$ divides $x^{p-1} - 1$ since

$$x^{p-1} - 1 = (x^q)^m - 1 = (x^q - 1)(1 + x^q + x^{2q} + \cdots + x^{(m-1)q}) = (x^q - 1)g(x), \quad (29)$$

where $g(x)$ is of degree $(m - 1)q = (p - 1) - q$. By Fermat's little theorem, the polynomial $x^{p-1} - 1$ has $p - 1$ distinct roots modulo p (any $x \in (\mathbb{Z}/p\mathbb{Z})^*$ is a root). So $x^q - 1$ must have q distinct roots modulo p , and $g(x)$ has $(p - 1) - q$ distinct roots. On the other hand, since

$$x^q - 1 = (x - 1)(1 + x + x^2 + \cdots + x^{q-1}) = (x - 1)\Phi_q(x) \quad (30)$$

the cyclotomic polynomial must have $q - 1$ distinct roots modulo p .